

Instructions:

- **Format:** Please submit a PDF file with typed answers (L^AT_EX or otherwise). Handwritten notes will not be graded.
- **Submission:** Only homework uploaded to Gradescope will be graded. Make sure to show all the steps of your derivations in order to receive full credit.
- **Integrity and Collaboration:** You are expected to work on the homework by yourself. You are not permitted to discuss them with anyone except the instructor. The homework that you hand in should be entirely your own work. You may be asked to demonstrate how you got any results that you report.
- **Clarifications:** If you have any questions, please look at Piazza first. Other students may have encountered the same problem, and is solved already. If not, post your question there. We will respond as soon as possible.

[10 points] Let X and Y be discrete random variables with mean 0, variance 1, and covariance ρ . Show that $\mathbb{E}(\max\{X^2, Y^2\}) \leq 1 + \sqrt{1 - \rho^2}$.

[20 points] Consider a sequence $\{X_n, n = 1, 2, 3, \dots\}$ such that,

$$X_n = \begin{cases} n & \text{with probability } \frac{1}{n^2} \\ 0 & \text{with probability } 1 - \frac{1}{n^2} \end{cases}$$

Show that,

1. $X_n \xrightarrow{p} 0$
2. $X_n \xrightarrow{L_r} 0$, for $r < 2$
3. X_n does not converge to 0 in the r -th mean for $r \geq 2$
4. $X_n \xrightarrow{a.s.} 0$

[20 points] Let X, Y , and Z be three jointly continuous random variables with joint PDF,

$$f_{XYZ}(x, y, z) = \begin{cases} x + y & 0 \leq x, y, z \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Find the joint PDF of X and Y . Find the joint PDF of X and Y .
2. Find the marginal PDF of X .
3. Find the conditional PDF of $f_{XY|Z}(x, y|z)$
4. Are X and Y independent of Z ?

[10 points] Let $X_1, \dots, X_n \sim P$ and let $P_n(A) = \frac{1}{n} \sum_{i=1}^n I(X_i \in A)$, where I is an indicator function which takes a value 1 when $X \in A$ and 0 otherwise. Let $\Delta_n = \sup_A |P_n(A) - P(A)|$. Show that $P(|\Delta_n - \mathbb{E}(\Delta_n)| > \epsilon) \leq 2e^{-2n\epsilon^2}$.

[20 points] In this problem, we will prove the optimality of the minimum mean square error estimator that is commonly used in machine learning.

1. Prove that $\mathbb{E}(Yg(X)) = \mathbb{E}[\mathbb{E}(Y|X)g(X)]$ if X and Y are jointly continuous random variables.
2. The above implies that $\mathbb{E}((Y - \mathbb{E}(Y|X))g(X)) = 0$. In other words, the estimation error $Y - \mathbb{E}(Y|X)$ is independent of every function of X . Use this fact to prove that if $\mathbb{E}(Y) < \infty$, then for any measurable function g ,

$$\mathbb{E}[(Y - \mathbb{E}(Y|X))^2] \leq \mathbb{E}[(Y - g(X))^2]$$

So this theorem shows that the error for the predictor $\mathbb{E}(Y|X)$ is lower than that of any other predictor $g(X)$. The optimal predictor $\mathbb{E}(Y|X)$ is called the Bayes estimator.